

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Bachelor of Technology (Mechanical Engineering) SEMESTER - 3 Summer 2025 ( Supply. )**

**Course : Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology**

**Semester : SEMESTER - 3**

**Subject Code & Name: BTMES304 - MATERIALS SCIENCE AND METALLURGY**

**Time : 3 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

**Q1. Objective type questions. (Compulsory Question)**

**12**

- 1 Following metal(s) possess(es) HCP crystal structure: (a) Zinc (b) Titanium(c) Cobalt(d) All of these
- Cracks are ----- defects. (a) 0-D (b) 1-D (c) 2-D (d) 3-D
- The number of favorable slip systems is highest in ----- crystal structures.(a) BCC (b) FCC (c) HCP (d) None of these
- is the ability of the material to absorb shocks.(a) Toughness(b) Stiffness (c) Hardness (d) None of these
- Tension test is used for evaluating the properties of ----- metals.(a) Ductile (b) Brittle (c) Both 'a' and 'b' (d) None of these
- hardness test gives more accurate readings.(a) Brinell (b) Rockwell (c) Both 'a' and 'b' (d) None of these
- Lower critical temperature for steel is ----- degree Celsius.(a) 210 (b) 727 (c) 910 (d) 1147
- Following test is used for the evaluation of hardenability:(a) Brinell (b) Vickers (c) Rockwell (d) Jominy
- Normalizing process involves cooling of the object in -----.(a) air (b) oil (c) water (d) furnace
- Following surface hardening treatment is used for depositing both C and N into the components' surface:(a) Carburizing (b) Nitriding (c) Carbonitriding (d) None of these
- Dye penetrant test is useful for detection of flaws in -----materials.(a) Ferrous (b) Nonferrous (c) Ceramics(d) All of these
- Following is/are the strengthening mechanism(s): (a) Refinement of grain size (b) Solid solution strengthening(c) Cold working (d) All of these

**Q2. Solve the following.**

- A) What are the three basic types of crystal structures? Explain any two of those w.r.t. schematic diagram, and atomic packing factor. **6**
- B) Classify the crystal defects and explain any two using schematic diagrams. **6**

- Q3. Solve the following.
- Compare the Brinell hardness test with Rockwell hardness test with the help of schematic diagrams. 6
  - Draw a neat diagram for Fe-Fe<sub>3</sub>C equilibrium diagram. Show all the phases, transformations and important temperatures on it. 6
- Q4. Solve Any Two of the following.
- Classify steels and explain any two types in detail. 6
  - Discuss the following heat treatment processes in detail: (a) Annealing (b) Normalizing (c) Hardening. 6
  - Describe the following surface hardening processes in detail: (a) Carburizing (b) Nitriding (c) Induction hardening. 6
- Q5. Solve Any Two of the following.
- Explain the procedure of specimen preparation for microscopy. Compare macroscopy with microscopy. 6
  - Explain principle and working of optical metallurgical microscope. 6
  - Describe the spark test w.r.t. purpose, mechanism of generation of spark, and the spark patterns for low carbon and high carbon steel. 6
- Q6. Solve Any Two of the following.
- What are the various strengthening mechanisms? Explain any two in detail. 6
  - Explain dye penetrant test with the help of schematic diagrams. 6
  - Describe magnetic particle inspection and ultrasonic test in brief. 6

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